Supervised Bayesian Matrix Factorization Identifies Prognostic Gene Expression Programs

Project Update

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August 27, 2026

Agenda

Background · The PDAC prognostic problem · why supervise the factorization

Method updates · The joint model & explicit priors · non-parametric baseline hazard · the ELBO & variational family · CAVI updates & convergence · preprocessing · selecting K: ARD shrinkage + a consensus of fit/prediction metrics

Results · The K_{init} consensus sweep · gene programs & survival · joint model vs. the unsupervised two-step, including as survival signal strength varies · multi-cohort simulation under ARD-based K selection

Impact & Future Directions

Roadmap for the next ~15-20 minutes. K selection works in two stages: a consensus of ELBO, BIC, log-likelihood, and cross-validated external C-index picks K_init, and ARD shrinkage then determines which of those factors are actually survival-active (K_eff). The projection predictor (YF) is the only model presented.

Background

Background: PDAC needs molecular prognosis

- **Five-year survival $< 12\%$** — among the most lethal solid tumors
- Most patients diagnosed late; **no widely adopted molecular stratification** at diagnosis
- Bulk expression mixes malignant, stromal, and immune signals — prognostic structure is interwoven and often **low-variance**
- **Goal:** recover *gene expression programs (GEPs)* whose activation predicts survival, and validate them on **independent** cohorts
- A GEP is a coordinated set of genes (a column of F) whose patient-level activation (a column of L) we test for survival association

Why model expression and survival *jointly*?

The common pipeline is **two-step**: run unsupervised factor analysis (PCA / EBMF), then test factors for survival *post hoc*.

- Unsupervised axes track the **largest expression variance** — often batch or housekeeping signal
- A program explaining little variance but strongly prognostic gets **missed**
- **Supervision** steers factors toward survival-relevant structure *during* estimation

A program explaining 15% of variance may be pure batch effect; one explaining 3% may cleanly separate survivors. Supervision prioritizes the second.

One background slide. The clinical motivation is unchanged. The methodological motivation — joint over two-step — is the thread of the whole talk, and I back it with both a synthetic comparison and a real-data external comparison against the unsupervised two-step (EBMF → Cox) baseline.

Methods

The model — two coupled components

$$\underbrace{Y = LF^\top + E, \quad E_{ij} \sim N(0, \tau_j^{-1})}_{\text{genomics: low-rank+gene-specific noise}} \qquad \underbrace{h_i(t) = h_0(t)\exp(\eta_i)}_{\text{Cox proportional hazards}}$$

Quantity	Dim	Meaning
Y	$n \times p$	expression matrix
L	$n \times K$	patient program-activation scores
F	$p \times K$	gene weights per program
β	$K \times 1$	survival (log-HR) coefficients
τ_j	scalar	gene-specific noise precision
η_i	scalar	patient log-risk score, $(Y_i F)\beta$

Shared F couples the two

The **same** gene-program weights F drive both the low-rank reconstruction of Y **and** the survival risk score $\eta = (YF)\beta$. A program is **prognostic only if its** $\beta_k \neq 0$; otherwise it explains expression alone.

Non-parametric baseline hazard

$h_0(t)$ is left **unspecified** — it cancels in the Cox partial likelihood. No Weibull / exponential shape assumption; only proportional hazards.

Two equations, one shared latent structure. Every symbol is defined in the table, including the risk score eta, which is now always the projection score (YF) . The baseline hazard is non-parametric, so h0(t) drops out of the partial likelihood and we make no parametric shape assumption.

The model: projection predictor and explicit priors

Projection predictor — $\eta = (YF)\beta$

- Risk uses the **direct genomic projection** YF, not a re-derived patient score
- **Exact** out-of-sample scoring: $\eta_{\text{new}} = (Y_{\text{new}}F)\beta$ — the *identical* formula at train and test, no train/test score mismatch
- L is fit for the genomics reconstruction only; it does not enter the survival likelihood

> Why the projection form
>
> Leakage-free **exact** external scoring, with no OLS re-projection step for new patients — the property that makes external validation on held-out cohorts a clean apples-to-apples comparison.

Priors — every block is explicit. $q(\cdot)$ denotes the **variational (approximate posterior)** for each block, fit by empirical Bayes at every update:

- $q(L_{\cdot k})$, $q(F_{\cdot k})$ — **point-exponential** (non-negative spike-and-slab): $g(\theta) = \pi_0\delta_0 + (1-\pi_0)\,\text{Exp}(\lambda),\ \theta \geq 0$
- $q(\beta_k)$ — **normal** (Gaussian slab, no spike): $g(\beta) = N(0, \sigma^2)$
- $q(\tau_j)$ — **closed-form MLE**: no EBNM, no prior tuning

Sparsity π_0, scale λ, and slab variance σ^2 are all estimated from the data (`ebnm`) at each step. The **normal prior on** β, not a spike-and-slab, is what makes ARD-based K selection possible (next: *selecting* K) — shrinkage toward zero, not a hard on/off switch.

The projection predictor (YF) scores new patients with the identical formula used at train time — no mismatch, no re-derivation. Priors are explicit: non-negative point-exponential on L and F, a Gaussian (normal) prior on , and a closed-form MLE for . The normal prior on   is worth flagging here because it's exactly the mechanism ARD-based K selection relies on next.

The objective: ELBO & variational family

Mean-field variational family — each block factorizes independently:

$$q(L, F, \beta, \tau) = \prod_{k=1}^{K} q(L_{\cdot k}) \prod_{k=1}^{K} q(F_{\cdot k}) \prod_{k=1}^{K} q(\beta_k) \prod_{j=1}^{p} q(\tau_j)$$

The **ELBO** (evidence lower bound) is the objective CAVI maximizes — a lower bound on the log marginal likelihood. For the projection model:

$$\text{ELBO} = \underbrace{E_q[\mathcal{L}_{\text{gen}}]}_{\text{expression fit}} + \underbrace{\lambda\, E_q[\mathcal{L}_{\text{Cox}}]}_{\text{survival fit}} - \underbrace{\textstyle\sum_{\text{blocks}} D_{\text{KL}}(q \,\|\, \text{prior})}_{\text{prior regularization}}$$

- **Genomics fit** $E_q[\mathcal{L}_{\text{gen}}] = \sum_j E_q \log p(Y_{\cdot j} \mid L, F_j, \tau_j)$ — Gaussian reconstruction of Y by $LF^\top$
- **Survival fit** $E_q[\mathcal{L}_{\text{Cox}})]$ — a **2nd-order Taylor** expansion of the Cox partial log-likelihood in $\eta = (YF)\beta$, making the survival term conjugate to the EBNM updates

- **KL term** penalizes each posterior q away from its prior — the adaptive shrinkage
- The **shared** F enters both likelihood terms — genomics reconstruction ($Y \approx LF^\top$) and survival risk ($\eta = (YF)\beta$) — that coupling *is* the supervision

- The ELBO **balances** expression and survival automatically — **no tuning parameter** α ($\lambda = 1$ in the recommended fits)

  What the ELBO is *for*: it is the **fitting objective** — CAVI maximizes it, and a fit is declared converged when it stops improving. It is also **one of four criteria** — alongside BIC, log-likelihood, and cross-validated external C-index — used to choose K_{init} (next: *Selecting* K).

The variational family is mean-field and factor-wise. The ELBO has three pieces: Gaussian genomics reconstruction, the Cox survival term via a second-order Taylor expansion that keeps the updates conjugate, and a per-block KL penalty that is the adaptive shrinkage. The shared F in both likelihood terms is the supervision. The ELBO is one of the four K_init-selection criteria alongside BIC, log-likelihood, and cross-validated external C-index.

Inference: factor-wise CAVI

Coordinate ascent (Gauss–Seidel) — per outer iteration, for each factor k:

$$\beta_k \;\to\; L_{\cdot k} \;\to\; F_{\cdot k}, \qquad \text{then } \tau \text{ once per sweep}$$

- Each update is an **EBNM** sub-problem: pseudo-data $x = B/A$, scale $s = 1/\sqrt{A}$, then `ebnm` returns the posterior mean **and** variance
- β_k has precision governed by the projection score's own variance across patients, so a factor with **no survival signal** shrinks $\beta_k \to 0$ on its own — the mechanism ARD relies on
- The empirical-Bayes prior $\hat{g}$ is **re-estimated every update** $\to$ adaptive shrinkage per factor / per gene; deflate the residual after each factor

Convergence — in plain English. After a 5-iteration burn-in, stop when the **ELBO stops improving** — its relative change between iterations falls below 10^{-5}:

$$\frac{\left|\text{ELBO}^{(t)} - \text{ELBO}^{(t-1)}\right|}{\left|\text{ELBO}^{(t-1)}\right|} < 10^{-5}$$

- The ELBO is the **single objective** being maximized, so this directly detects a stationary point of the variational optimization
- Mean absolute changes in L and β are tracked alongside as **diagnostics** (both settle below 10^{-3} at convergence)

! Risk-direction sign correction

Cox concordance is invariant to a global sign flip, so an unconstrained fit can land at $C < 0.5$. **Fix:** orient each program against the **training** concordance, then apply the *same* orientation at test. (Used to define program activation in the survival results.)

CAVI updates one factor at a time in Gauss-Seidel order; each block reduces to an EBNM call with pseudo-data x=B/A and scale s=1/sqrt(A). Convergence requires both L and beta to settle below 1e-3 in MEAN absolute change; averaging avoids a single oscillating factor blocking the stop. The sign correction orients each program by training concordance and reuses it at test.

Preprocessing is crucial: per-platform normalization

The $p \gg n$ scaling problem. Merging platforms (RNA-seq, microarray, proteomics) leaves each gene on a different scale. The survival precision for program k is

$$A_k = \sum_i w_i \, (YF)^2_{ik},$$

where w_i = the patient's **Cox working weight** (curvature from the survival Taylor step) and $(YF)_{ik}$ = patient i's **projection score** on program k. Without correction, between-platform variance inflates (YF) unevenly, **swamps** A_k, and the survival coefficients **collapse to** $\beta \to 0$.

The fix — normalize *per platform, before merging*:

1. Log2 transform (RNA-seq only)
2. **Per-cohort gene selection** (combined rank of mean × variance), top-3000 per cohort, then **intersect** → **2064 genes** in the merged training matrix
3. **Per-platform z-standardization** (each cohort centered/scaled on its own)
4. Row-bind into the merged training matrix

! Skip per-platform normalization → collapses

In an 18-configuration benchmark, **10 of 12** pipelines that do **not** z-standardize per platform before merging drive $\beta \to 0$.

No leakage

Normalization is fit **within each training cohort, before** any split; held-out cohorts get the *same* per-cohort recipe at scoring time.

The single most important practical finding since April. A__k is the survival precision for program k. Without per-platform normalization the projection scales blow up across platforms, A__k is dominated by between-platform variance, and beta collapses. The fix is per-platform z-standardization before merging, which leaves 2064 genes. 10 of 12 alternatives collapse. Normalization is fit per training cohort before any split — no leakage.

Selecting the number of programs: a two-stage framework

Stage 1 — K_{init}, by a consensus of four fit/prediction criteria. Fit at several K_{init}; compare ELBO, BIC, an ELBO-style joint log-likelihood, and cross-validated external C-index across the grid.

Stage 2 — the active-factor subset, by ARD. Fit once at the chosen K_{init}; the normal prior on β (previous slide) lets Automatic Relevance Determination shrink unneeded factors' $\beta_k \to 0$ **on its own**, without a second per-K validation loop.

Stage 1: choosing K_{init}

- **ELBO** — the fitting objective itself, compared *across* K_{init}
- **BIC** $= -2\,\text{LL} + \log(n) \cdot \text{df}$, $\text{df} = K_{\text{init}} \cdot (n + p + 1)$ — an honest complexity penalty; **not** guaranteed to agree with ELBO or external C
- **Log-likelihood** — genomics + survival, an ELBO-style bound (not an exact marginal likelihood)
- **Cross-validated external C-index** — the only *generalization* signal among the four, evaluated on the five held-out cohorts
- These need **not** agree — see *Results* for the full $K_{\text{init}} = 2..20$ sweep

Stage 2: ARD classification at fixed K_{init}

Each factor k is classified by `classify_factors()`:

- **survival-active**: $|\hat{\beta}_k| > 0.001$

- **genomics-only**: not survival-active, but $\text{PVE}_k > 1\%$
- **dead**: neither — fully pruned

Running example: $K_{\text{init}} = 7$

7 starting factors → **2 survival-active** + **2 genomics-only** (4 kept, 3 pruned). Stable across a wide range of K_{init} — see *Results*.

Two stages. Stage 1 picks K_init by a consensus of ELBO, BIC, log-likelihood, and cross-validated external C-index — these don't have to agree, and the Results section shows the full sweep and where they disagree. Stage 2 is ARD: at a fixed K_init, the normal prior on beta lets survival-irrelevant factors shrink to beta=0 on their own, and classify_factors() splits the rest into survival-active (|beta|>0.001) vs genomics-only (PVE>1%) vs dead. K_init=7 is the running example throughout: 2 survival-active, 2 genomics-only, 4 kept total, 3 pruned — the same real-data configuration used throughout Results.

Results

Data — Synthetic Validation & PDAC Cohorts

Multi-cohort simulation

EBMF-grounded ground truth: shared factors (present in all studies) + study-specific factors; **shared factors carry the survival signal**, study-specific do not.

Scenarios

- *All Shared Factors* — every program is shared & prognostic
- *All Cohort-Specific Factors* — negative control; no shared signal ($\eta \equiv 0$)
- *Shared & Cohort-Specific Factors* — the realistic mix

Model Arms

- Unsupervised two-step (EBMF → Cox)
- The projection-predictor joint model, K selected by over-specified K_{init} + ARD pruning (mirroring the real-data procedure)

PDAC cohorts — External Validation

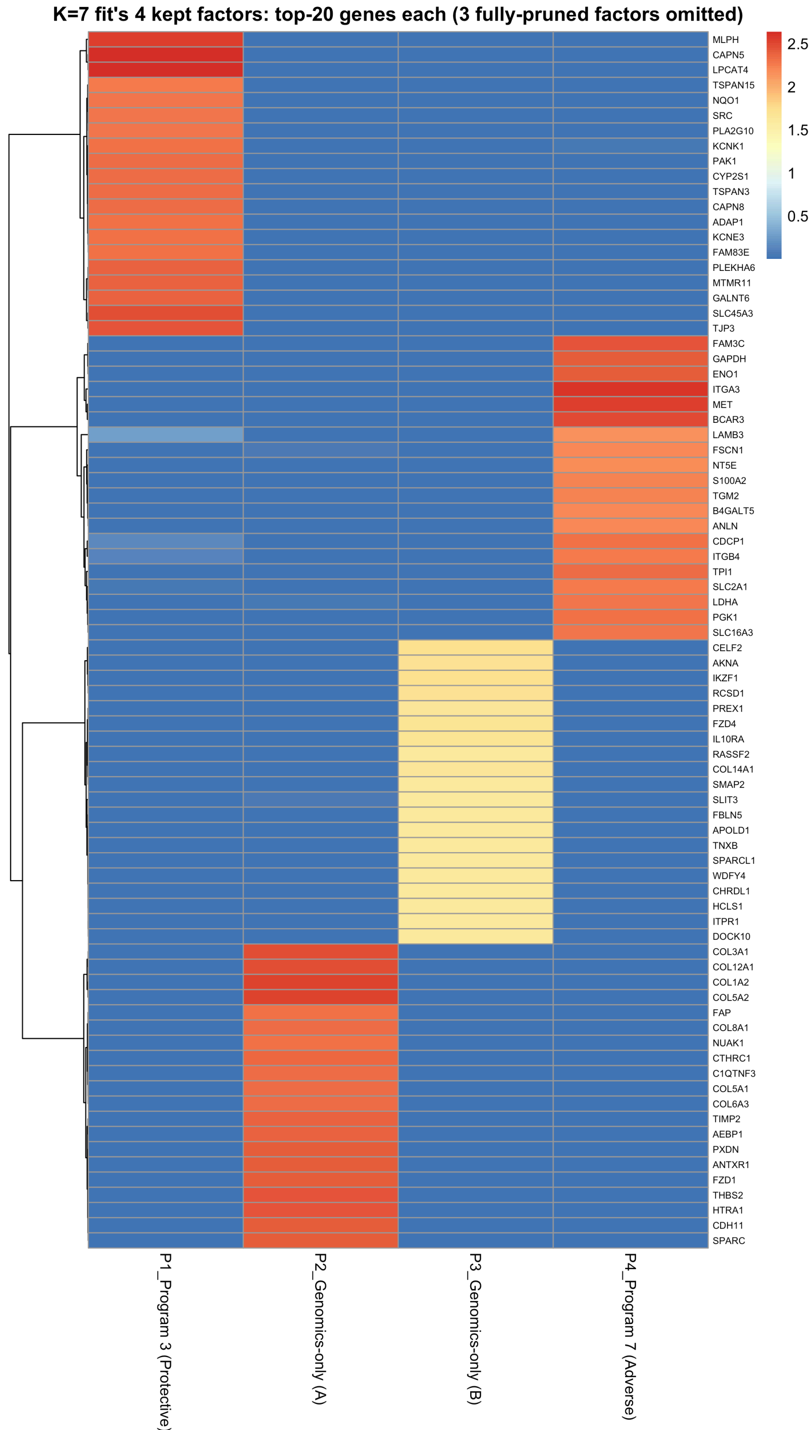
- **Train:** TCGA-PAAD + CPTAC ($n \approx 273$, RNA-seq + proteomics)
- **Validate:** 5 fully held-out cohorts, scored by the **exact** projection formula $\eta = (Y_{\text{new}} F)\beta$

Held-out cohort	Platform
Dijk	RNA-seq
Moffitt	Microarray
PACA-AU (array)	Microarray
PACA-AU (seq)	RNA-seq

Puleo	Microarray

Two datasets. The simulation now selects K the same way the real-data procedure does — an over-specified K_init pruned by ARD — instead of the oracle true-K used in June, and compares only the joint model against the unsupervised EBMF- then-Cox baseline. On real data we train on TCGA + CPTAC and validate on five fully held-out cohorts spanning microarray and RNA-seq, scored by the exact projection formula.

PDAC Cohorts — Prognostic Gene Programs



- $K_{\text{init}} = 7$ (ELBO / BIC / log-likelihood / external-C consensus — see the sweep below) → **2 survival-active** + 2 genomics-only kept — Program 3 (**protective**) and Program 7 (**adverse**)
- The 2 genomics-only programs are now biologically characterized too: **Program 5** reads as **desmoplastic stroma** (collagens, SPARC, THBS2, FAP), **Program 6** as **immune infiltrate** (IKZF1, AKNA, IL10RA, HCLS1) — real expression structure, distinct from the survival-active programs and from each other, just not prognostic on its own

On real data, the recommended fit starts at K_init=7 and ARD keeps 4 programs — 2 survival-active (Program 3 protective, Program 7 adverse) and 2 genomics-only (Program 5 stroma, Program 6 immune infiltrate). The figure shows only the 4 kept factors, not all 7 — the 3 fully-pruned (dead) factors are omitted since they carry no real gene-program structure. Weights are non-negative because L and F use a point-exponential prior.

PDAC Cohorts — Prognostic Association

The two survival-active programs stratify patients (split at median **projection score** $(YF)_k$ — the quantity the model scores on):

- **Adverse** — **Program 7** (higher activation → shorter survival), log-rank $p = 0.00047$
  - top genes: *ITGA3, MET, BCAR3, FAM3C, GAPDH, ENO1, TPI1, PGK1*
- **Protective** — **Program 3** (higher activation → longer survival), log-rank $p = 0.0077$
  - top genes: *CAPN5, LPCAT4, MLPH, SLC45A3, TJP3, PLEKHA6, MTMR11, GALNT6*

Proportional-hazards assumption not violated (Schoenfeld test).

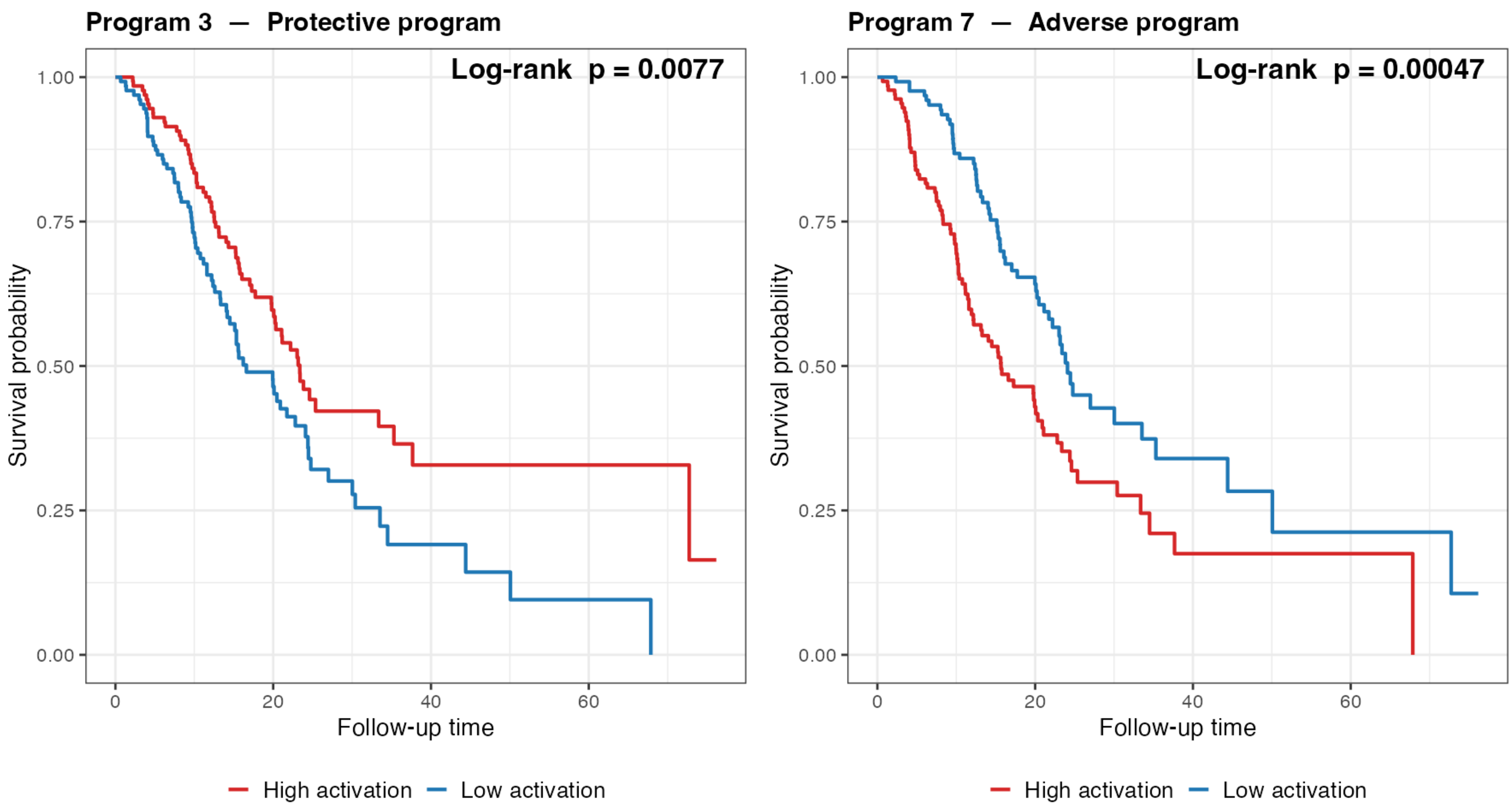


Figure 2: **Figure.** Kaplan–Meier curves for the two survival-active programs, patients split at the median projection score $(YF)_k$ in the training cohorts. The adverse program's high-activation group has worse survival; the protective program's, better. Direction read empirically from the curves.

The two active programs stratify patients into clear survival groups, split by the projection score (YF)__k. Program 7, the adverse program, carries MET, ITGA3, and glycolytic genes — biologically a classic aggressive signature. Program 3, protective, carries epithelial differentiation markers. Directions are read off the curves. Schoenfeld test shows PH holds.

Choosing K_{init}: the consensus sweep

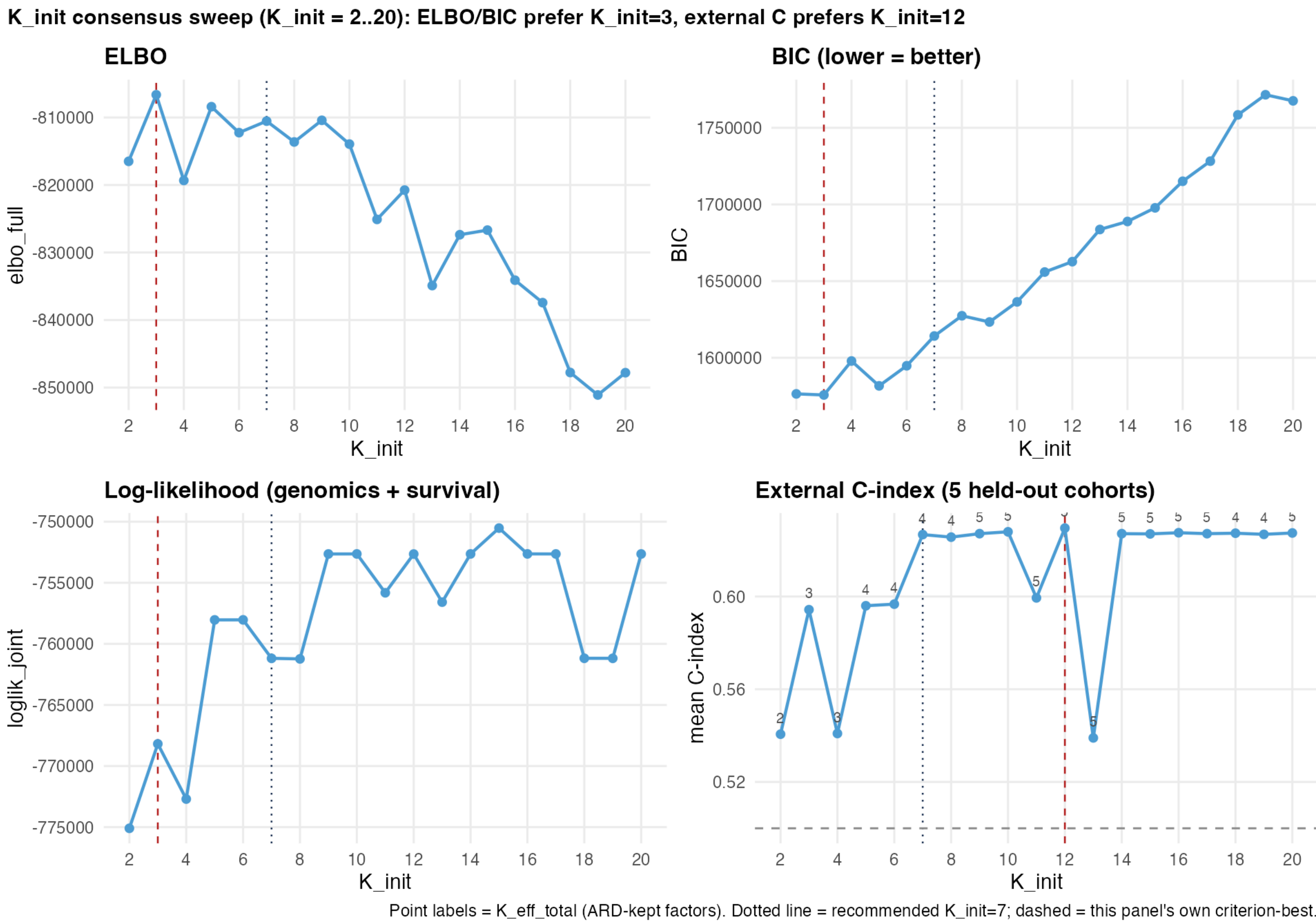


Figure 3: **Figure.** $K_{\text{init}} = 2..20$, each fit classified by ARD. Point labels on the C-index panel are K_{eff}total (ARD-kept factors). Dotted line = recommended $K_{\text{init}} = 7$; dashed = each panel's own criterion-best.

- **ELBO- and BIC-preferred:** $K_{\text{init}} = 3$ · **External-C-preferred:** $K_{\text{init}} = 12$ — these do **not** agree, reported honestly, not retuned to force a match
- **Recommendation:** $K_{\textbf{init}} = 7$**.** Survival-active count is stable at 2 across a wide range of K_{init}; $K_{\text{init}} = 7$ is externally competitive (on the C-index plateau) and the simplest single-fit path — full table in the appendix

Here is where the disagreement matters. ELBO and BIC agree with each other, preferring K_init=3 – BIC's df penalty, tens of thousands of units per K_init here, dominates once K_init climbs into double digits. External C-index prefers K_init=12, but K_init=7 through 20 mostly sit on a plateau around C~0.627, with two isolated dips at K_init=11 and 13 – I want to show the actual curves and provoke the question of whether a different K_init in that plateau might be preferable, rather than presenting K_init=7 as the only defensible choice. K_init=7 is kept as the practical recommendation because the survival-active count is stable there and it's the simplest single-fit path.

Joint model vs. the unsupervised two-step

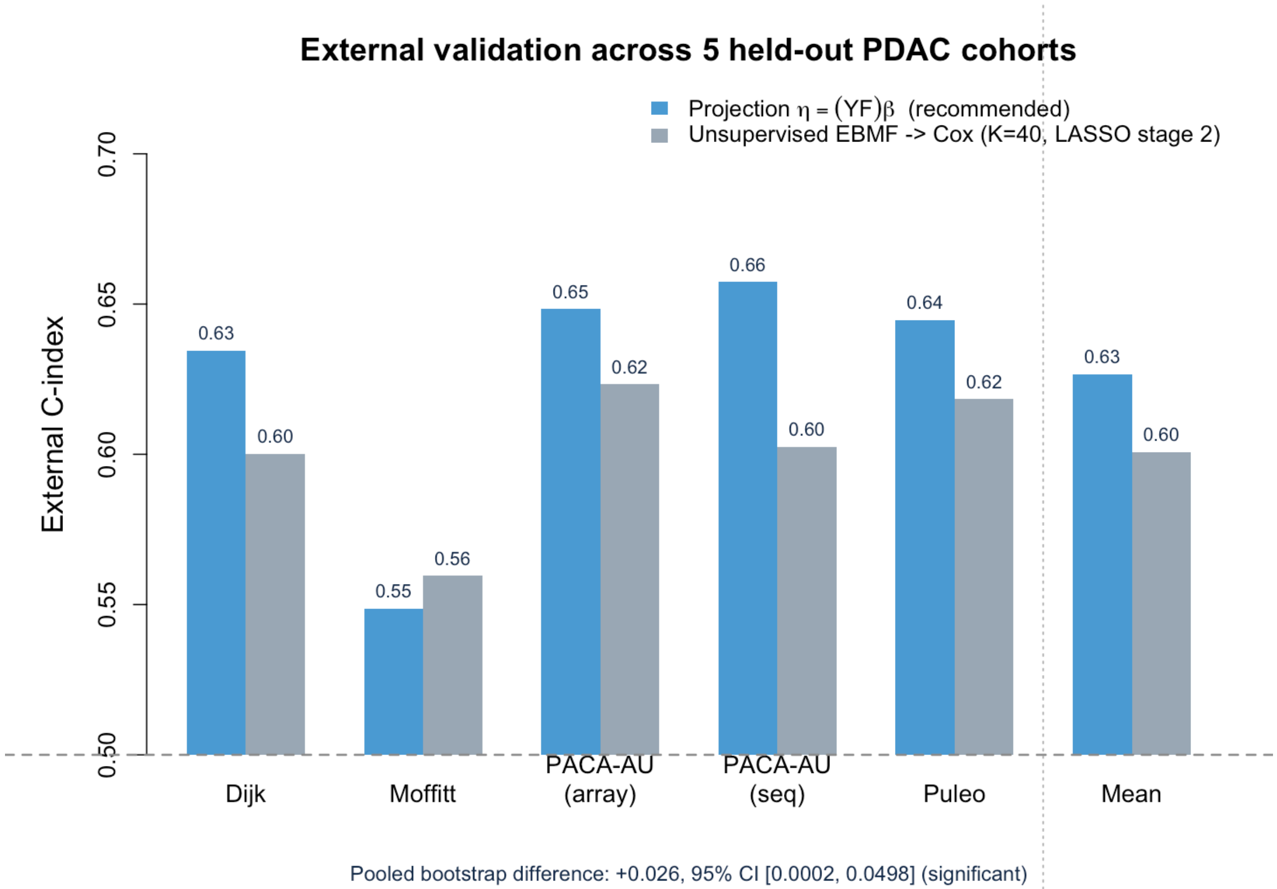


Figure 4: **Figure.** External C-index across the five held-out PDAC cohorts (+ mean): the recommended projection model vs. the fairest independent two-step baseline (EBMF → Cox, $K = 40$, LASSO stage 2 — a large, YFB-uninformed factor count with a regularized second stage, rather than a small K or an unregularized `coxph()` that overfits at $K = 40$). Dashed line = chance.

- **Projection model mean C = 0.627** · unsupervised EBMF → Cox (K=40, LASSO) 0.601, across the five held-out cohorts

- Pooled bootstrap difference: **+0.026, 95% CI [0.0002, 0.0498]**, n=616 pooled across cohorts — **significant**
- $K_{\text{init}} = 7$ (ELBO / BIC / log-likelihood / external-C consensus) → $K_{\text{eff}} = 2$ survival-active of 4 kept

The real-data analog of the synthetic slide. Two arms: the recommended projection model, and the fairest independent two-step baseline I could build – EBMF at a large, YFB-uninformed K=40, with a regularized (LASSO) stage-2 Cox rather than a plain coxph() that overfits at that K. The pooled bootstrap difference is significant. This is not the only two-step comparison in the project – other configurations trail YFB by more, some without reaching significance at this sample size – this is the one built specifically to be hardest to argue with.

Joint model vs. two-step as survival signal strength varies

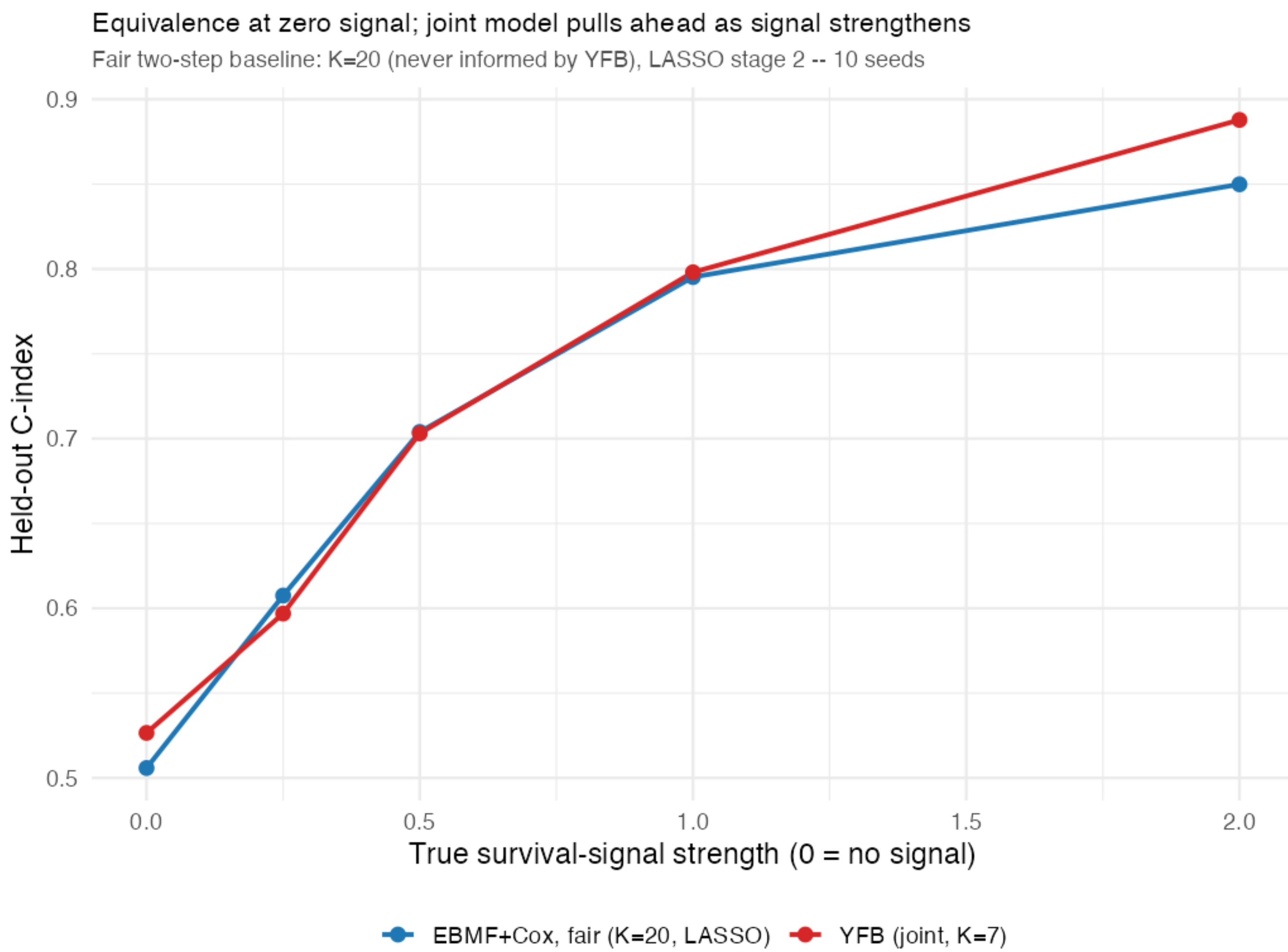


Figure 5: **Figure.** External performance of the joint model vs. the unsupervised two-step (EBMF → Cox) as the simulated survival-signal strength increases. At strength = 0 (no true survival signal in the data), the joint model matches EBMF — supervision costs nothing when there is nothing to exploit. At strength = 2, the joint model pulls ahead.

- **Strength = 0:** joint ≈ EBMF-only — the joint model does not fabricate prognostic signal when none exists
- **Strength = 2:** joint model outperforms the two-step — supervision helps *only when there is signal to find*, and does not hurt when there isn't

This single figure carries two points. At zero simulated survival signal, the joint model tracks EBMF almost exactly — no false discovery, no manufactured prognostic signal. As signal strength increases, the joint model pulls ahead of the two-step baseline. Together this is the cleanest argument that supervision is a free option: it costs nothing when uninformative and helps when it isn't.

Synthetic Results — ARD matches oracle-K performance

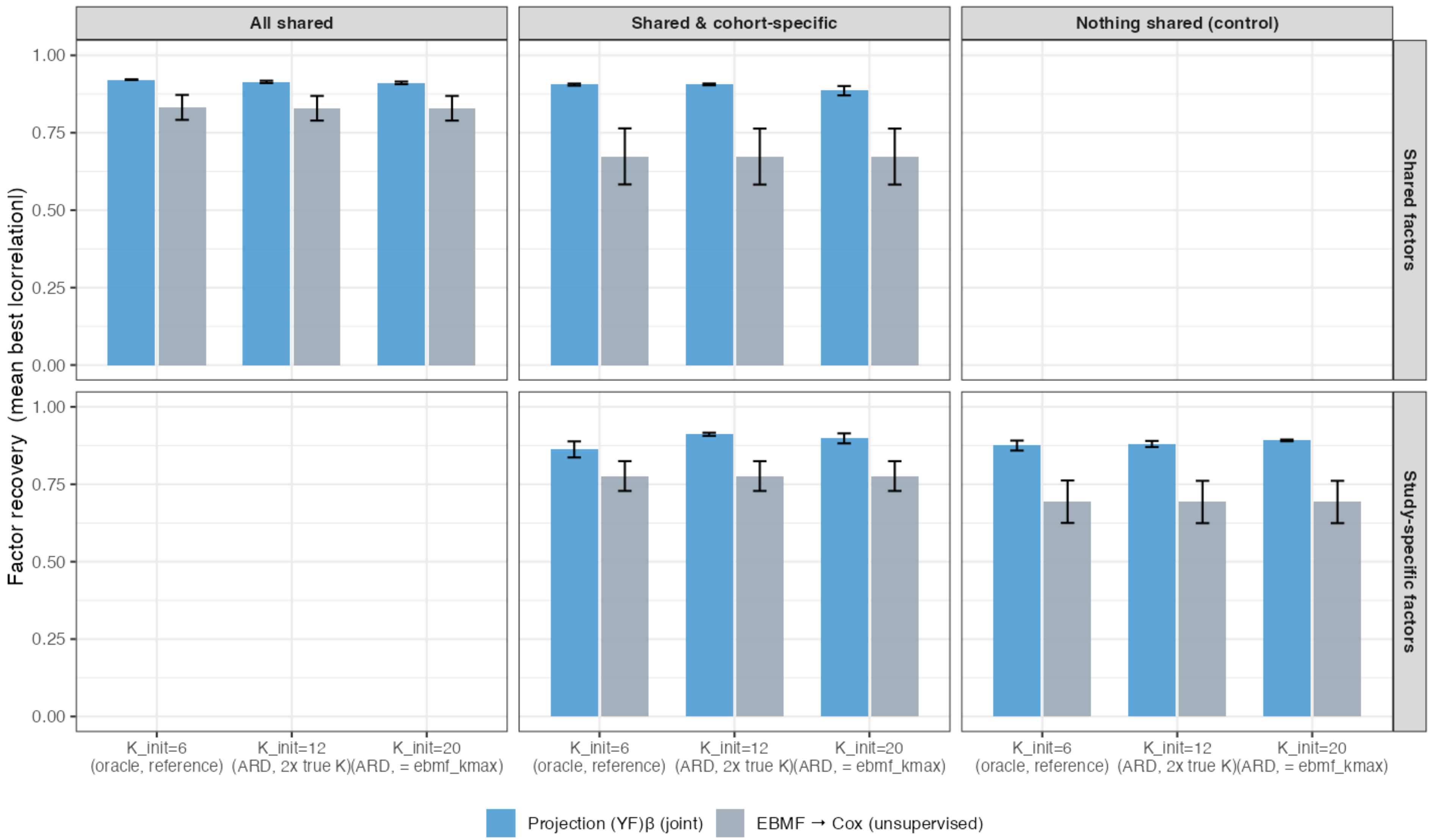


Figure 6: **Figure.** Factor recovery (mean best |correlation| vs. the true gene programs) by K_{init}, arm, and scenario (15 seeds).

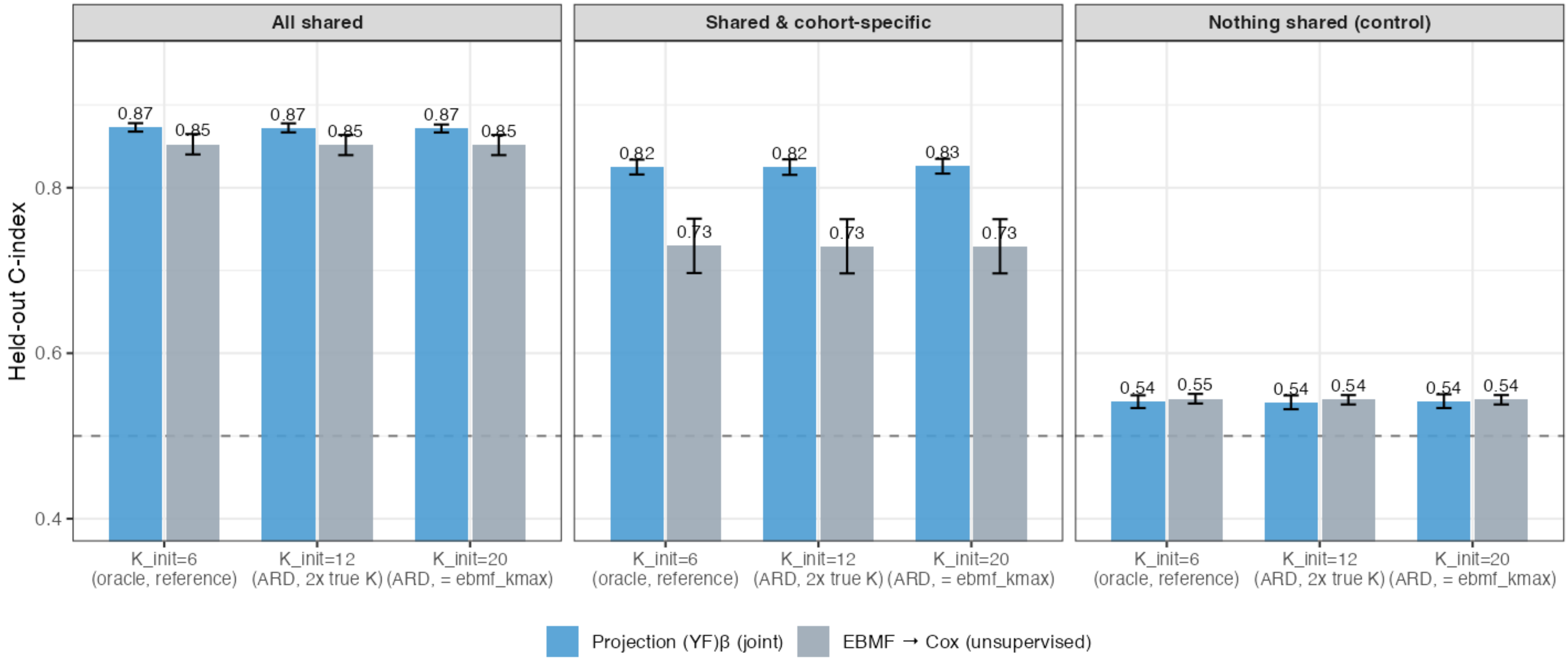


Figure 7: **Figure.** Held-out C-index by K_{init}, arm, and scenario (15 seeds; dashed = chance).

- Multi-cohort simulation now selects K the way the real-data procedure does — an over-specified $K_{\text{init}} \in \{12, 20\}$ pruned by ARD — instead of June's oracle true-$K = 6$; $K_{\text{init}} = 6$ retained as an internal reference
- **Recovery, specificity, and C-index are flat across** $K_{\text{init}} = 6 \to 12 \to 20$ in every scenario — over-specifying K and letting ARD prune costs nothing relative to knowing the true K, now with 15 seeds (up from 5) for a properly powered comparison

The simulation now uses the real-data K-selection procedure – over-specify K_init, let ARD prune – instead of June's oracle true-K, with K_init=6 retained as an internal reference point. The headline is a clean null result: recovery, specificity, and C-index don't move at all across K_init=6/12/20, in any of the three scenarios, with 15 seeds. Over-specifying K and trusting ARD costs nothing.

Synthetic Results — false-positive rate: a trend, not yet significant

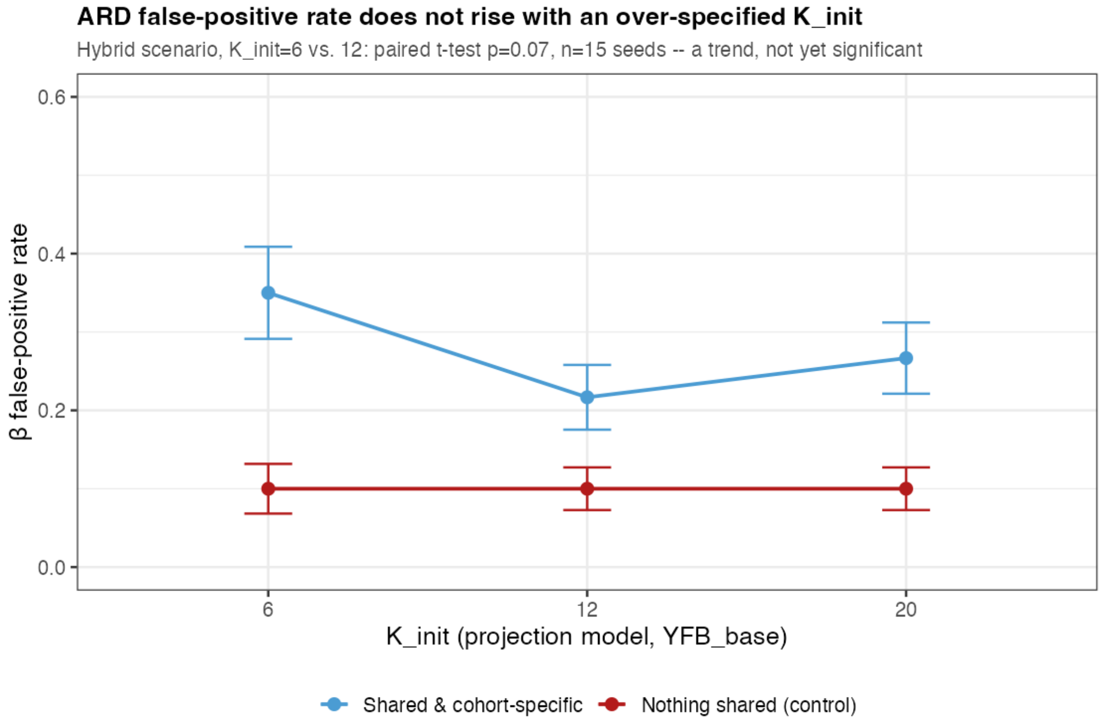


Figure 8: **Figure.** β false-positive rate (a non-signal factor falsely called survival-active) vs. K_{init}, YFB_base only (EBMF has no survival coefficient to test). 15 seeds.

- In the realistic **hybrid** scenario, the false-positive rate is numerically lower at $K_{\text{init}} = 12/20$ than at the oracle $K_{\text{init}} = 6$ ($0.35 \to 0.22 \to 0.27$) — but a paired test (same seeds, $K_{\text{init}} = 6$ vs. 12) gives $p = 0.07$: **directionally consistent, not conventionally significant** at $n = 15$ seeds
- In the **negative-control** scenario, the false-positive rate is flat at 0.10 across all K_{init} — no change either direction
- Reported as an honest trend worth watching, not a confirmed effect — and consistent with the advisor discussion that a small, non-zero coefficient on an extra factor contributes little to the risk score in practice

This is the one place a genuinely new question could have shown up – does over-specifying K produce MORE false-positive survival-active calls, since ARD has more non-signal factors to potentially mis-activate. At 5 seeds it looked like a clear improvement; at 15 seeds the direction holds in the hybrid scenario but the paired test is p=0.07, not significant, and the negative control is completely flat. I'm presenting this as what it is – a promising trend, not a confirmed finding – and tying it back to the point raised after the last simulation exercise: a small spurious coefficient on a non-signal factor doesn't meaningfully move the risk score, which is consistent with false positives here not being large in magnitude even when they occur.

Model Comparison Summary

Model	Linear predictor	Preprocessing
Projection (recommended)	**(YF)**	**per-platform z-std + survival-ra… selection**
Unsupervised two-step	EBMF → Cox	per-platform z-std + survival-ranked g…

Real ext. C = mean across 5 held-out PDAC cohorts. **K_init** = starting factor count, selected by an ELBO / BIC / log-likelihood / external-C consensus (Stage 1); **K_eff** = survival-active programs by ARD, $|\hat{\beta}| > 10^{-3}$ (Stage 2). Gene selection ranks genes by combined mean × variance per cohort.

- **Recommendation:** projection predictor $(YF)\beta$ · per-platform z-standardization · survival-ranked gene selection · **no** cohort indicator
- $K_{\text{init}} = 7 \to K_{\text{eff}} = 2$ survival-active (4 kept total), with **exact** external scoring

> **i** On gene selection
>
> The combined-rank selection recipe was **motivated by** the lab's penalized survival-MF (DeSurv) preprocessing; here it is applied within the fully Bayesian model.

Two rows: the recommended projection model against the unsupervised two-step baseline. Projection predictor with per-platform z-std and survival-ranked gene selection, K_init=7 selected by consensus, is the recommendation.

Impact / key findings

1. **Joint supervision beats the unsupervised two-step**, and costs nothing when there is no survival signal to exploit (signal-strength sweep).
2. **Preprocessing is decisive** — per-platform z-standardization before merging is the difference between a working model and $\beta \to 0$; **10 of 12** alternatives collapse.
3. K **selection is a two-stage split** — cross-validated external C-index (with ELBO, BIC, and log-likelihood) helps pick K_{init}; ARD shrinkage then determines the survival-active subset ($K_{\text{eff}} = 2$) directly from that single over-specified fit, with no separate per-K validation loop needed for K_{eff} itself.

4. **Over-specifying K_{init} and trusting ARD costs nothing** — confirmed in simulation (15 seeds): recovery, specificity, and C-index are flat whether the model starts at the true K or 3x over-specified.
5. **Mean external C-index = 0.627** across five independent PDAC cohorts — fully Bayesian throughout (adaptive priors, no α tuning, uncertainty-aware).

Limitations

- **BIC, ELBO, and external C-index need not agree** on K_{init} — reported honestly, not retuned to force a match (previous slides)
- **The ARD retention thresholds (PVE $> 1\%$, $|\hat{\beta}| > 0.001$) are not literature-derived** — $\hat{\beta}$'s cutoff was reverse-engineered from this model's own coefficient scale, not a principled sparse-factor cutoff; a sensitivity sweep and literature check are open (ROADMAP.md)
- **Small external cohorts give noisy estimates** — per-cohort C ranges down to 0.55 (Moffitt), near chance
- The active programs are **statistically** prognostic and **biologically annotated** (pathway enrichment: Program 7 basal-like/adverse, Program 3 classical/protective, confirmed by 5 independent methods) — a matched head-to-head against DeSurv on one protocol remains a planned next step

Five takeaways, then an honest limitations beat: the K-selection criteria disagree with each other and that's reported as a finding, not smoothed over; small external cohorts give noisy per-cohort estimates; and the biological annotation of the two active programs is done, with a matched DeSurv comparison still on the to-do list.

Future directions & summary

Current status

- **K selection**: cross-validated external C-index, ELBO, BIC, and log-likelihood inform K_{init}; ARD shrinkage determines K_{eff}
- **Gene programs biologically annotated** — pathway enrichment on both survival-active programs, 5 independent methods

Next steps

1. **Multiple cohorts / multiple modalities via indicator columns in L** — stack methylation + expression, possibly with modality-specific priors; leave-one-study-out validation; train on A vs. B vs. A+B
2. **Manuscript preparation** — methods-primary framing; PDAC as the application

Summary

1. Joint supervised factorization via the projection predictor $(YF)\beta$; **point-exponential priors on L, F, a normal prior on** β, non-parametric baseline hazard
2. **Preprocessing is crucial** — per-platform normalization prevents the survival precision from being dominated and β from collapsing
3. **K selection: cross-validated external C-index, ELBO, BIC, and log-likelihood inform K_{init}; ARD shrinkage determines K_{eff}** — validated on real data (five external PDAC cohorts, mean C = 0.627) and synthetic data with known ground truth
4. **Recommended model**
   - Linear predictor: projection $(YF)\beta$
   - L cohort indicator: No (candidate direction for multi-cohort/multi-modality extension, above)
   - Pre-processing: per-platform z-std
   - $K_{\text{init}} = 7 \rightarrow K_{\text{eff}} = 2$, mean external C = **0.627**

K selection: cross-validated external C-index, ELBO, BIC, and log-likelihood inform K_init; ARD shrinkage determines K_eff. Both survival-active programs are now biologically annotated. The forward-looking direction I want to lead with is extending L with indicator columns for multiple cohorts and/or multiple modalities — stack methylation and expression, possibly with modality-specific priors, and validate with leave-one-study-out and train-on-A-vs-B-vs-A+B comparisons. Then manuscript prep.

Discussion

Thank You

Questions, suggestions, and feedback welcome!

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Open the floor. Appendix slides cover convergence diagnostics, survival-activity magnitudes, and the DeSurv comparison.

Appendix

Appendix — K_{init} sweep, full table

K_init	Surv-active	Genomics-only	Dead	K_eff (kept)	ELBO	BIC	Log-lik	External C
2	1	1	0	2	-816,495	1,576,416	-775,093	0.5406
3	1	2	0	3	-806,636	1,575,712	-768,183	0.5943
4	1	2	1	3	-819,328	1,597,855	-772,698	0.5409
5	3	1	1	4	-808,404	1,581,651	-758,038	0.5960
6	3	1	2	4	-812,248	1,594,760	-758,035	0.5967
7	**2**	**2**	**3**	**4**	**-810,540**	**1,614,174**	**-761,185**	**0.6267**
8	2	2	4	4	-813,629	1,627,379	-761,230	0.6256
9	2	3	4	5	-810,421	1,623,321	-752,643	0.6271
10	2	3	5	5	-813,944	1,636,429	-752,640	0.6279
11	3	2	6	5	-825,096	1,655,903	-755,819	0.5994
12	2	3	7	5	-820,752	1,662,678	-752,649	0.6295
13	3	2	8	5	-834,897	1,683,664	-756,585	0.5390
14	2	3	9	5	-827,366	1,688,890	-752,641	0.6271
15	2	3	10	5	-826,682	1,697,778	-750,527	0.6270
16	2	3	11	5	-834,100	1,715,115	-752,638	0.6275
17	2	3	12	5	-837,425	1,728,238	-752,642	0.6271
18	2	2	14	4	-847,771	1,758,428	-761,180	0.6273
19	2	2	15	4	-851,099	1,771,549	-761,183	0.6268
20	2	3	15	5	-847,800	1,767,584	-752,643	0.6274

Highlighted row = the recommended $K_{\text{init}} = 7$. External C = mean C-index across the 5 held-out PDAC cohorts. BIC and Log-lik are ELBO-style bounds, not exact marginal-likelihood quantities.

The full sweep table, for anyone who wants the actual numbers rather than just the curves. Note the K_eff (kept) column is not monotone in K_init – it moves between 2 and 5 as ARD prunes differently at each starting point, and two rows (K_init=11, 13) show a visible external-C dip against their neighbors.

Appendix — convergence & survival-activity magnitudes

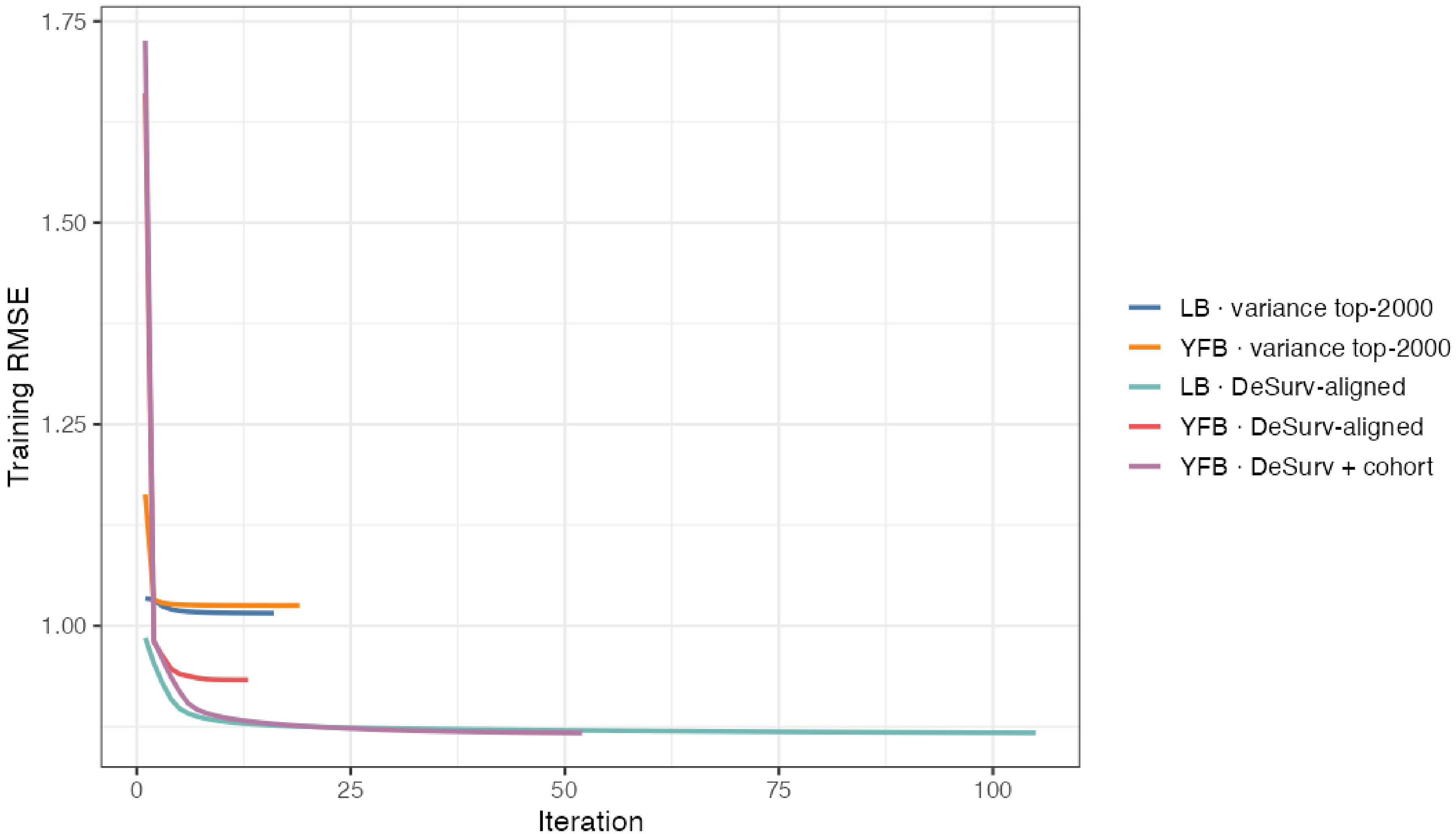

Figure 9: **Figure A1.** Training RMSE convergence traces across model configurations — RMSE stabilizes to a plateau; stopping is governed by the relative ELBO change falling below 10^{-5} ($\Delta L, \Delta\beta$ tracked as diagnostics).

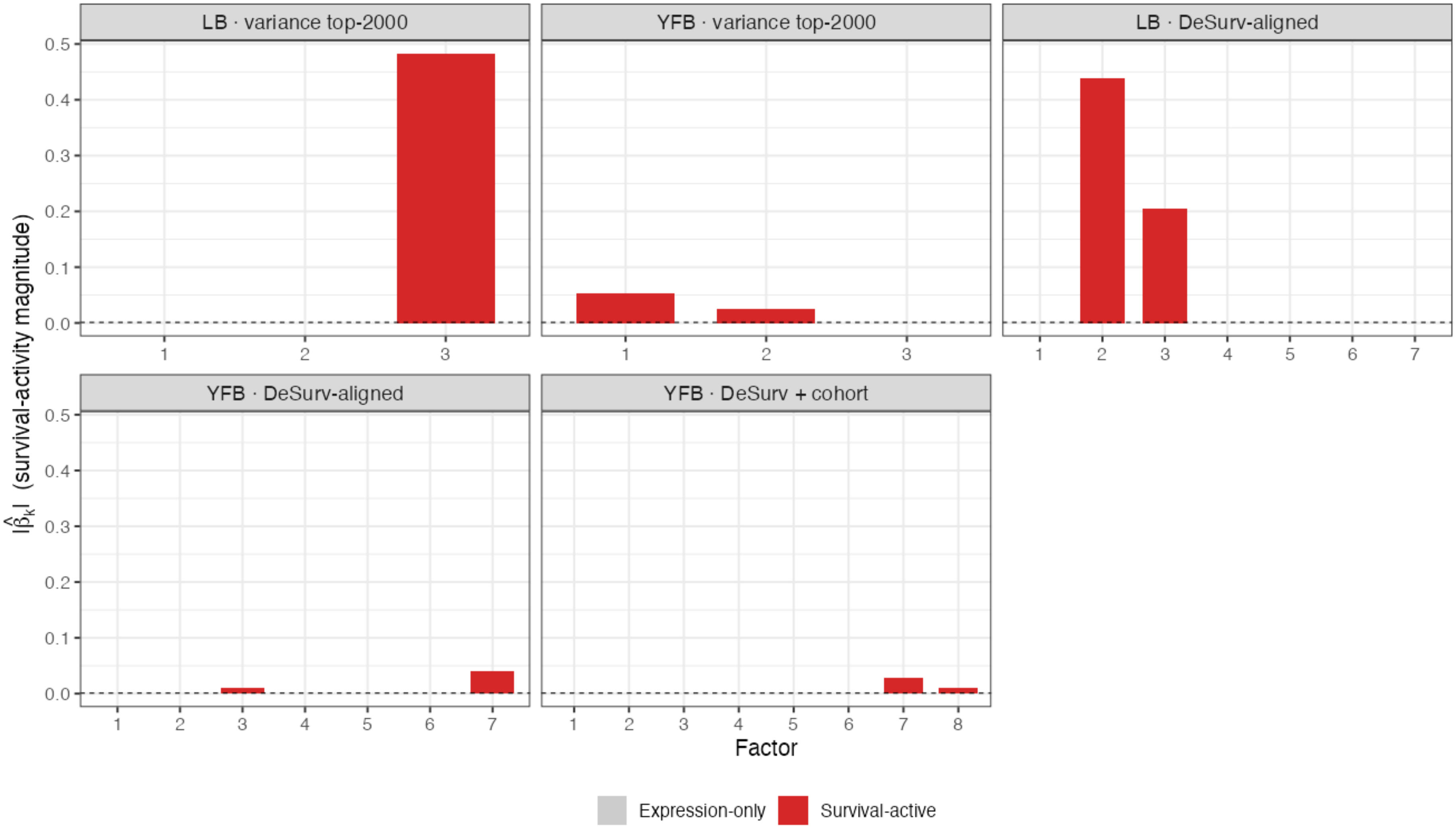

Figure 10: **Figure A2.** Survival-activity magnitude $|\hat{\beta}_k|$ per program. Most programs fall below the activity threshold (expression-only); a small number clear it (survival-active).

- The spike-and-slab / shrinkage behavior is visible: most programs are shrunk below the survival-activity threshold, with only the survival-active ones retained

Appendix — relation to DeSurv (Young et al., PNAS 2026)

Same design & same data — SBMF is the Bayesian counterpart. Both jointly fit supervised NMF + Cox with **non-negative gene programs**, share the same training set (**TCGA + CPTAC,** $n = 273$) and the **same five external PDAC cohorts** (Dijk, Moffitt, PACA-AU array/seq, Puleo; $n = 616$), score new patients by **projection** (no retraining), and use the **combined-rank top-3000-per-cohort** gene selection. Both also find that the **highest-variance program is *not* the most prognostic**.

DeSurv objective: $(1-\alpha)\,\mathcal{L}_{\mathrm{NMF}} - \alpha\,\mathcal{L}_{\mathrm{Cox}}$, with the survival gradient on the gene programs W.

What SBMF changes

	DeSurv	SBMF
Estimation	penalized optimization	Bayesian variational (CAVI)
K selection	Bayesian-optimization over k, α, λ jointly	ARD shrinkage + ELBO/BIC/log-likelihood/external-C consensus on K_{init} only
Shrinkage	fixed elastic-net	empirical-Bayes priors (learned)
Uncertainty	point estimates	posterior mean + variance
Noise	least-squares	per-gene precision τ; $E[L^2]$ correction

- **DeSurv:** $k = 3,\ \alpha = 0.334$; pooled external **HR/SD** = 1.50 (95% CI 1.31–1.72)
- **SBMF (YF) :** $K_{\mathrm{init}} = 7 \to K_{\mathrm{eff}} = 2$; mean external **C $=0.627**

Honest caveat: the two report different primary metrics (DeSurv: pooled HR/SD; SBMF: mean external C-index) and use slightly different normalization (DeSurv within-subject rank → 1970 genes; SBMF per-platform z-std → 2064). A **matched same-protocol head-to-head is a planned next step** — DeSurv is *not* re-run here.

This is the backup for the inevitable "how does this compare to DeSurv?" question. SBMF is the Bayesian sibling of DeSurv — same supervised NMF+Cox idea, same training data and the same five external cohorts, same projection-based scoring, and SBMF borrowed DeSurv's combined-rank gene selection. The genuine differences: DeSurv tunes k, alpha, and lambda jointly by Bayesian optimization; SBMF selects K_init by a consensus of fit/prediction metrics and lets ARD prune within that fit, with empirical-Bayes priors learning the shrinkage and no alpha or lambda to tune, plus posterior uncertainty and a heteroscedastic noise model. On numbers, DeSurv reports pooled HR per SD of 1.50; SBMF reports mean external C of 0.627 — different metrics, so a matched head-to-head on one protocol is the honest next step, not a claim of superiority today.